

User Manual: Safe Semantic Planner

PCCST503 - Machine Learning

1 Overview

This manual outlines how to configure, compile, and execute the C++ Safe Semantic Planner.

2 Compilation

The project requires a compiler supporting C++11 or higher. Compile the source utilizing standard g++:

```
g++ -std=c++11 planner.cpp -o planner
```

3 Execution and Configuration

- **Running the Planner:** Execute the compiled binary using `./planner`.
- **Defining the Problem:** Modify the `PlanningProblem` object in `main()`. You must define the `initialState`, `goalState`, and populate the `badStates` vector.
- **Modifying Transitions:** Transitions can be added or removed by updating the `transitions` vector. Each transition requires an ID, origin, destination, cost, safety score, reliability score, and availability flag.
- **Dynamic Replanning:** To simulate a dynamic environment, modify a transition's `available` flag to `false` and call `planner.plan(p)` again.